EXPERIMENT 7

Aim: Hands on Solidity Programming Assignments for creating Smart Contracts

Theory :

Solidity is a high-level programming language used to develop smart contracts on blockchain platforms such as Ethereum. A smart contract is a program stored on the blockchain that automatically executes predefined operations. Solidity supports variables, functions, control statements, data structures, modifiers, and constructors, which make it suitable for developing decentralized applications.

A Solidity program generally contains an **SPDX license identifier, pragma statement, and contract definition**. The `pragma` statement specifies the compiler version, while the contract contains state variables and functions. Solidity provides primitive data types such as `uint`, `int`, `bool`, `address`, `string`, and `bytes`. State variables are stored permanently on the blockchain, whereas local variables exist only during function execution.

Functions are used to perform operations in a smart contract. A `view` function can read contract state but cannot modify it, while a `pure` function neither reads nor modifies blockchain state. Functions can accept input parameters and return output values. Solidity also provides visibility levels such as **public, private, internal, and external** to control access to functions and variables.

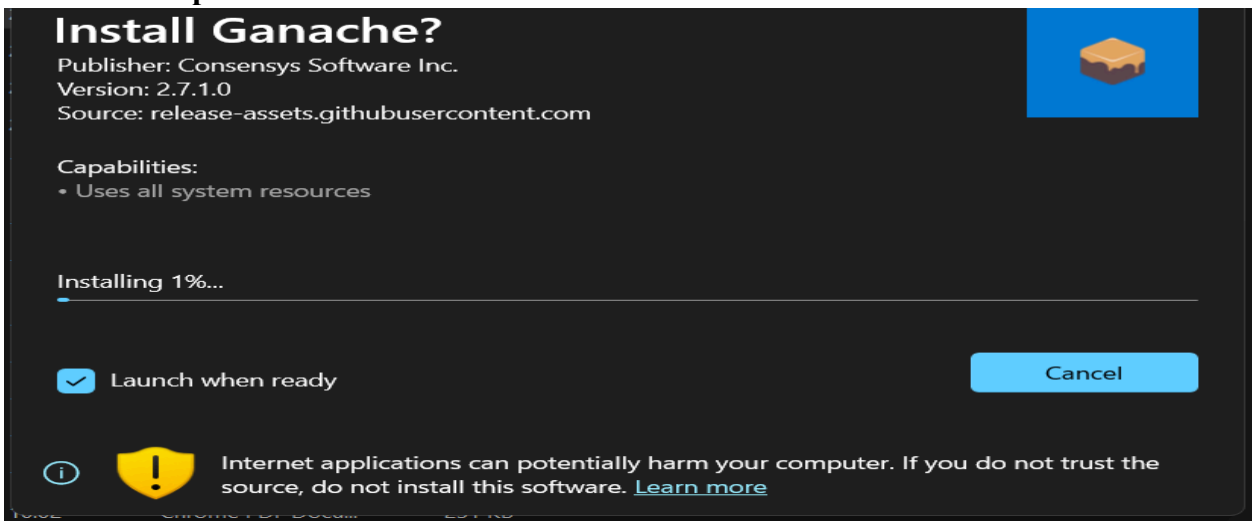
Modifiers are reusable code blocks used to control the execution of functions, commonly for access control. A **constructor** is executed only once during contract deployment and is generally used to initialize contract variables. Solidity also supports control-flow statements such as `if-else`, `for`, `while`, and `do-while` for implementing decisions and repeated operations.

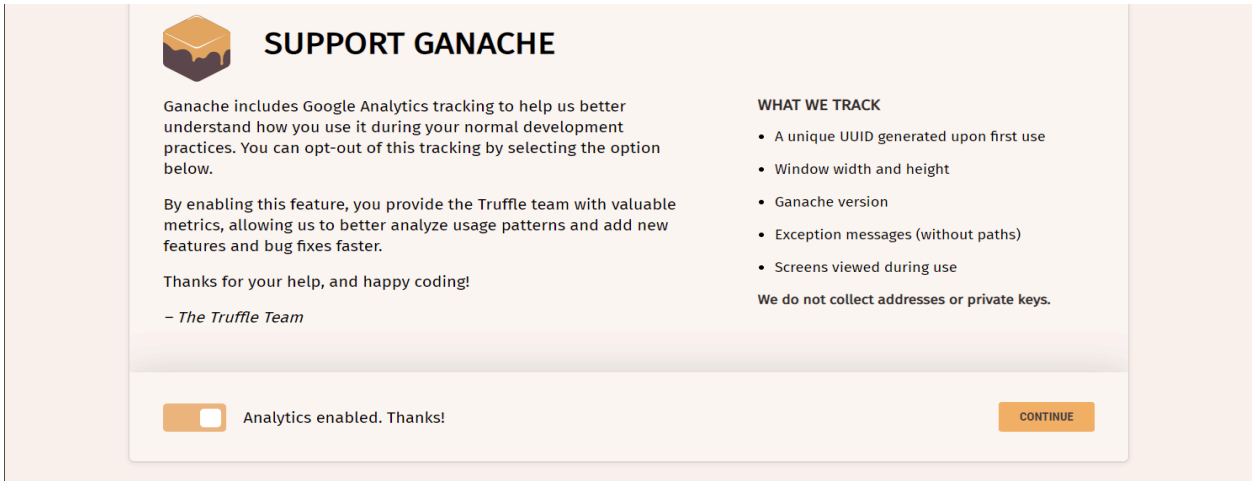
Solidity provides important **data structures** such as arrays, mappings, structures, and enums. Arrays store collections of elements, mappings store key-value pairs, structs group related data, and enums represent a predefined set of values. Solidity also uses three major data locations: **storage**, which stores data permanently on the blockchain; **memory**, which stores temporary data during function execution; and **calldata**, which is temporary and read-only data generally used for external function parameters.

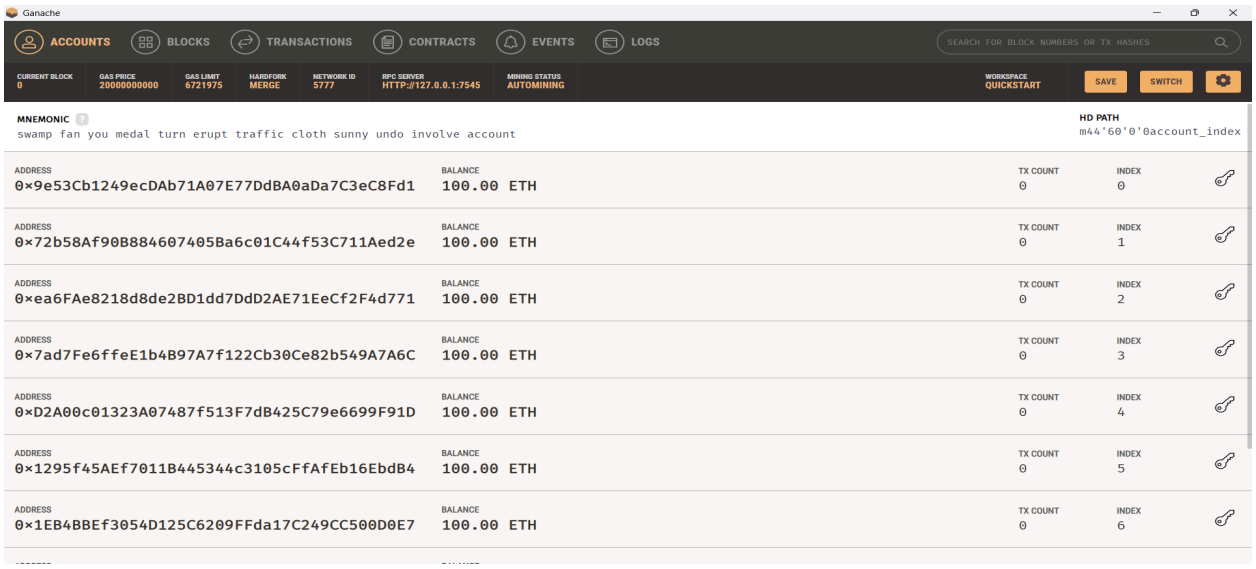
Smart contracts can also handle **Ether**, the native currency of Ethereum. Ether is represented internally in **Wei**, where **1 Ether = 10^{18} Wei**. Every state-changing transaction requires **gas**, which represents the computational effort needed to execute the operation. The gas price determines the cost paid for each unit of gas. Therefore, efficient smart contract design is important for reducing transaction costs.

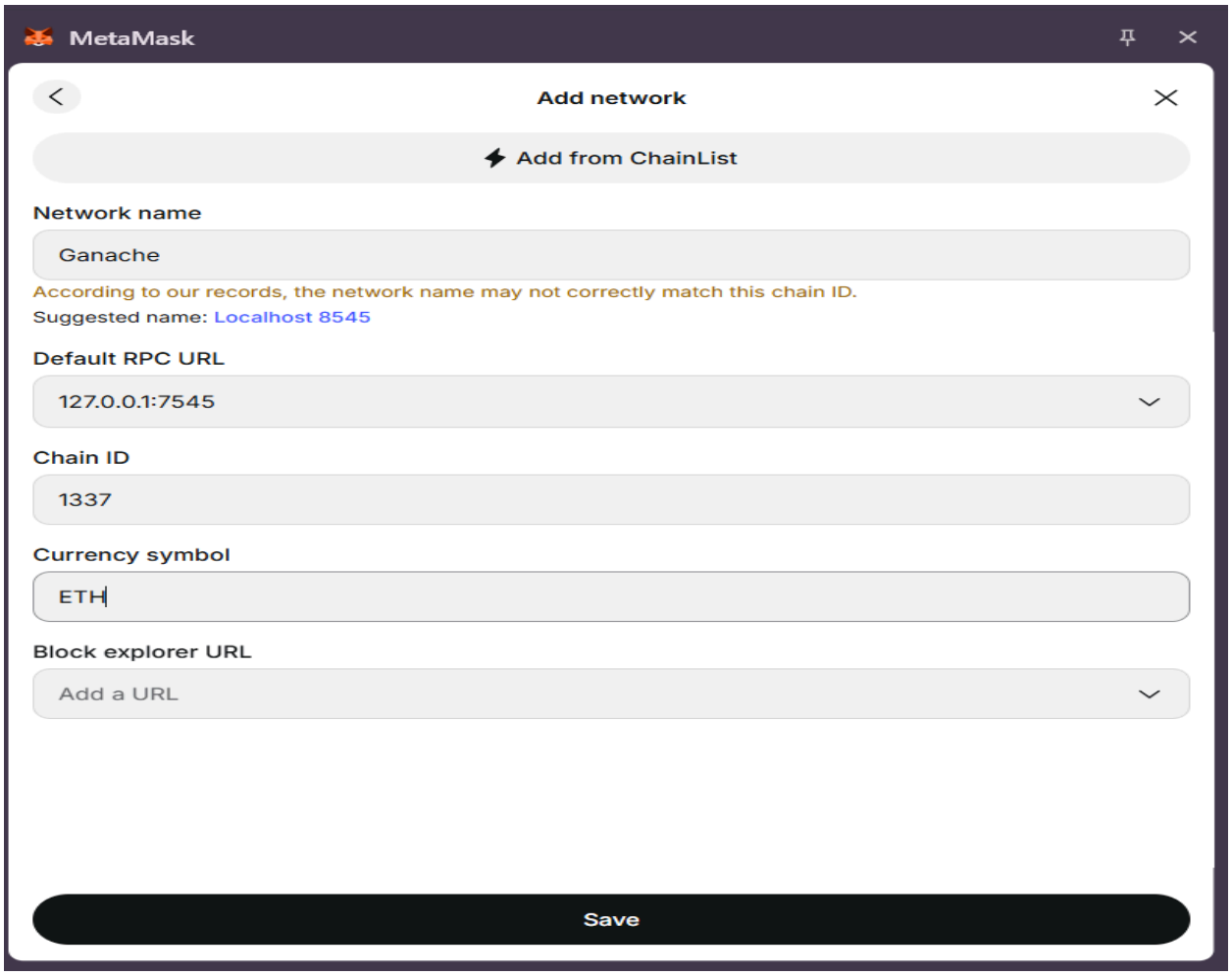
In this experiment, **Remix IDE** is used as the development and deployment environment. The Solidity programs are written, **compiled**, and then **deployed on the Remix VM** for testing. Compilation converts the Solidity source code into executable bytecode and generates the information required to interact with the contract. After deployment, contract functions can be called and their results can be observed through Remix.

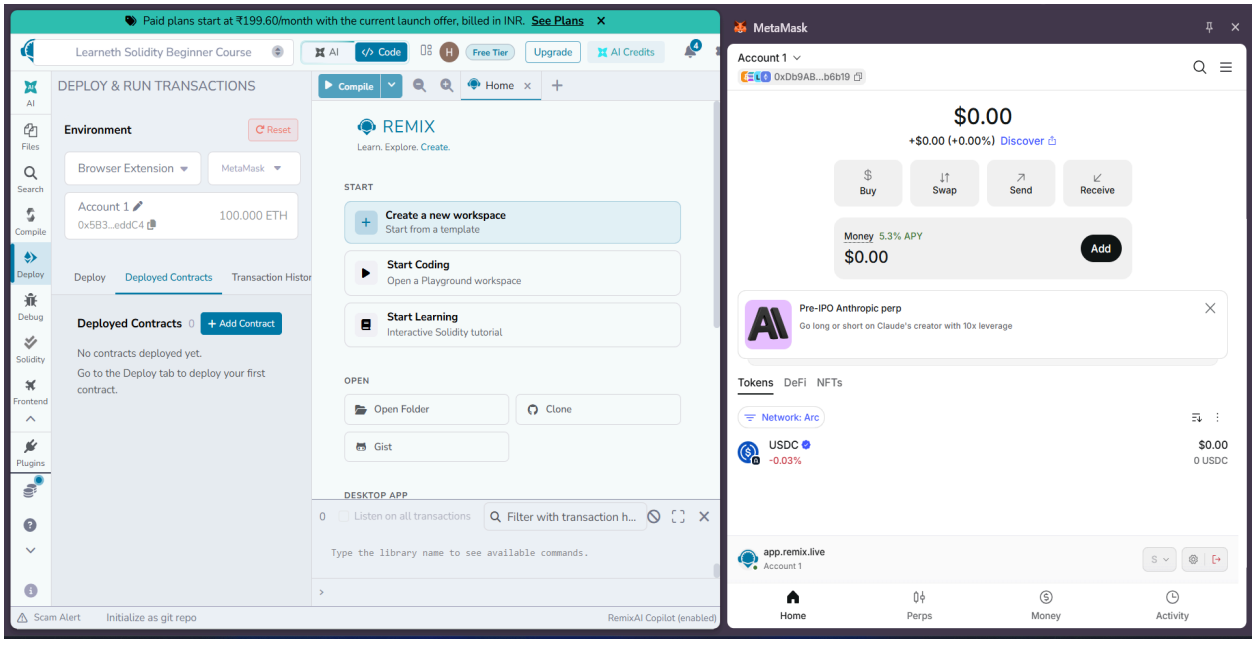
Code and Output :

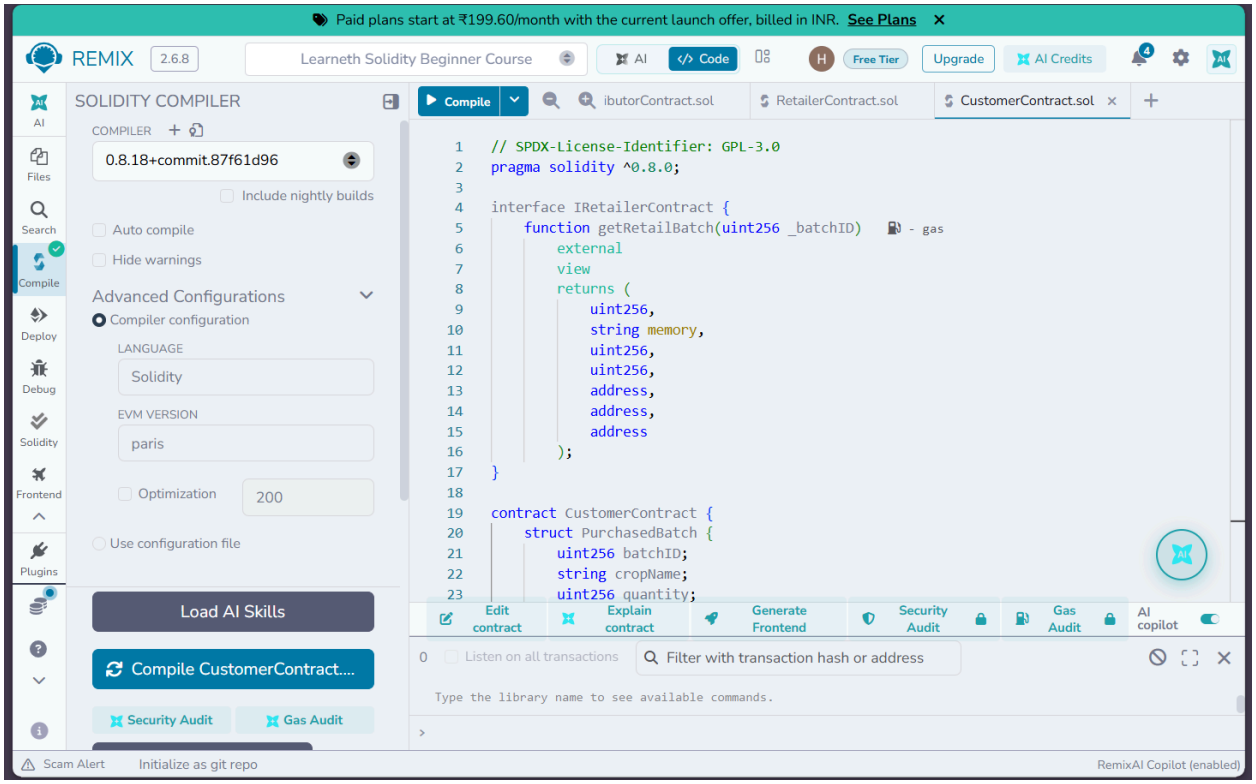


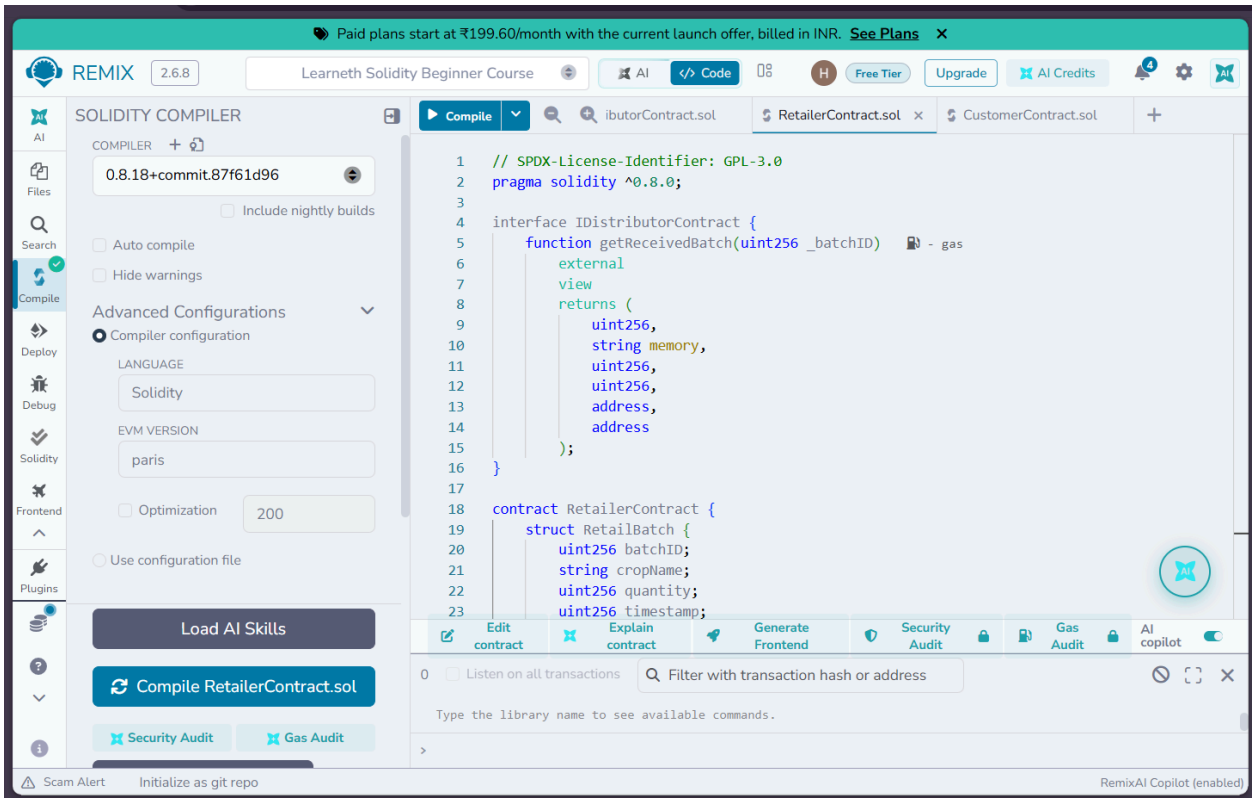


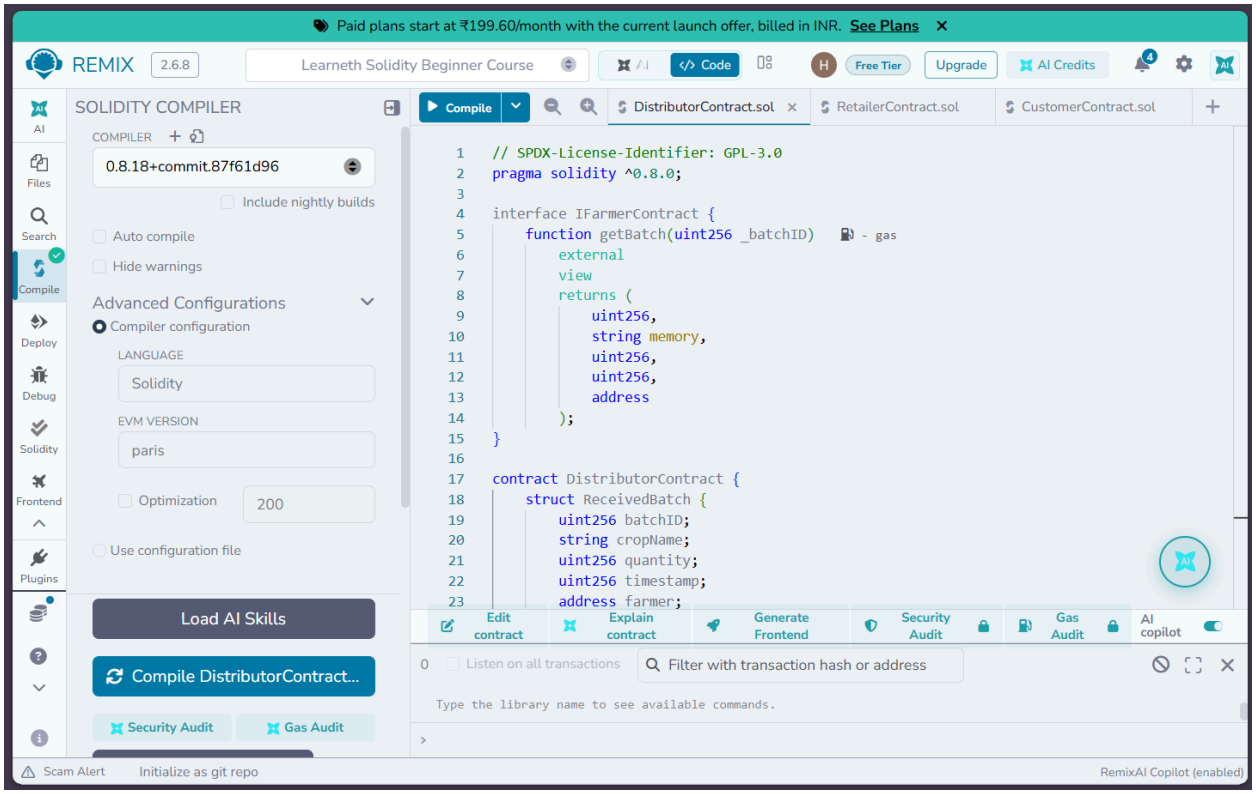


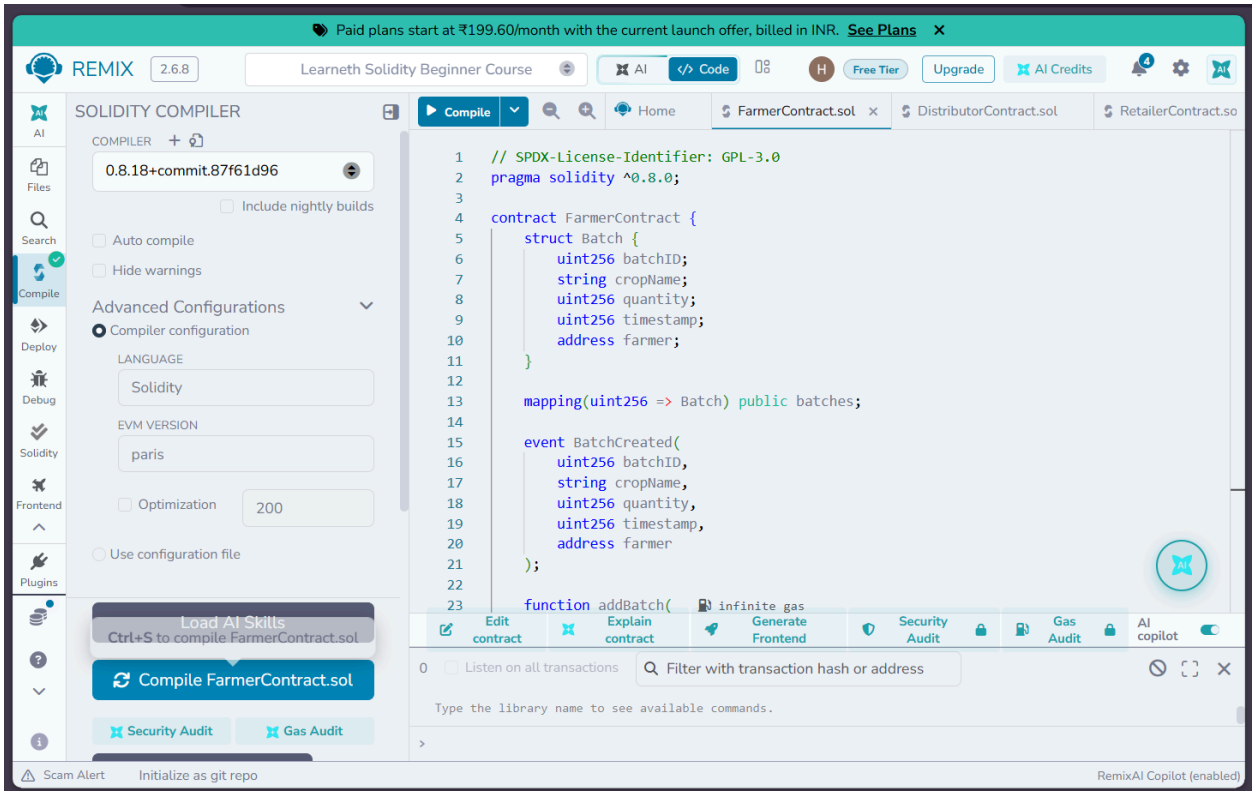

MetaMask
Add network
Add from ChainList
Network name
Ganache
According to our records, the network name may not correctly match this chain ID.
Suggested name: Localhost 8545
Default RPC URL
127.0.0.1:7545
Chain ID
1337
Currency symbol
ETH
Block explorer URL
Add a URL
Save

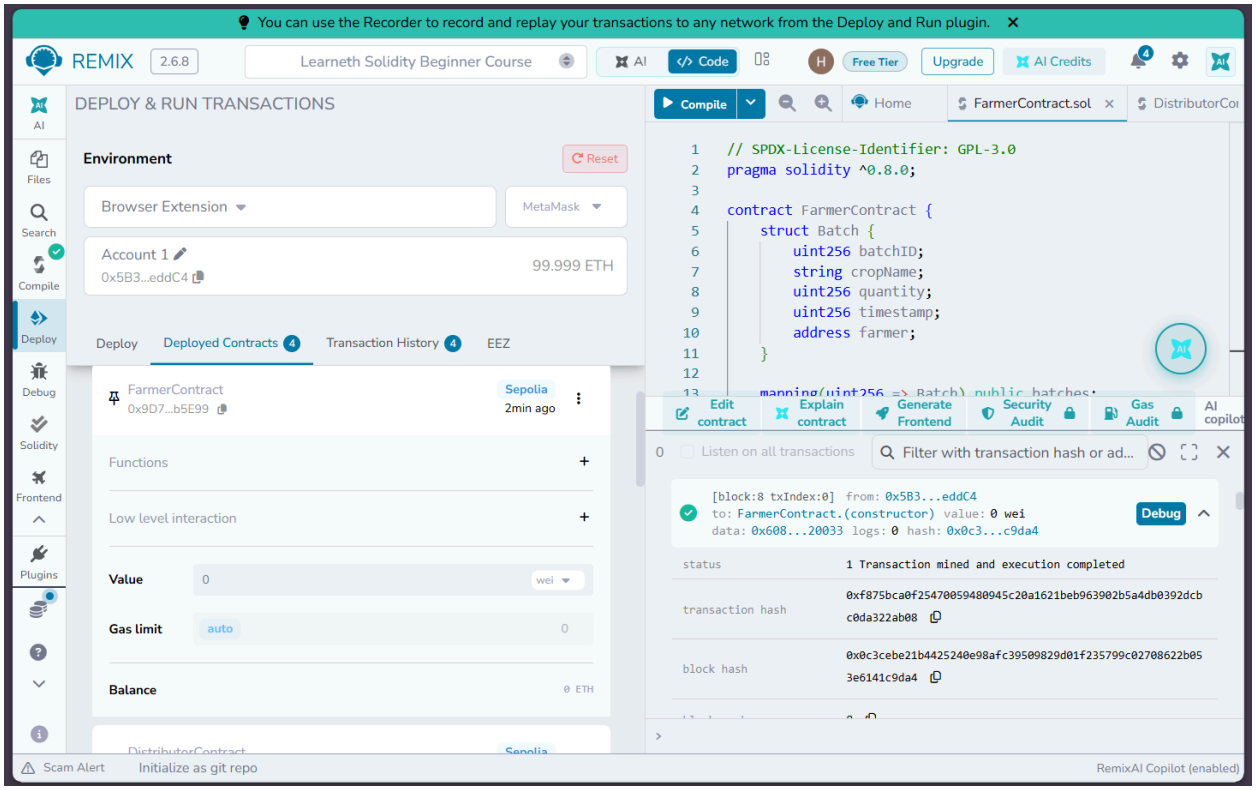

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Learneth Solidity Beginner Course
DEPLOY & RUN TRANSACTIONS
Environment
Browser Extension
MetaMask
Account 1
100.000 ETH
Deploy
Deployed Contracts
Transaction Histor
Deployed Contracts 0
+ Add Contract
No contracts deployed yet.
Go to the Deploy tab to deploy your first contract.
REMIX
Learn. Explore. Create.
START
Create a new workspace
Start from a template
Start Coding
Open a Playground workspace
Start Learning
Interactive Solidity tutorial
OPEN
Open Folder
Clone
Gist
DESKTOP APP
Type the library name to see available commands.
Scam Alert
Initialize as git repo
RemixAI Copilot (enabled)
MetaMask
Account 1
$0.00
+$0.00 (+0.00%) Discover
Buy
Swap
Send
Receive
Money 5.3% APY
$0.00
Add
Pre-IPO Anthropic perp
Go long or short on Claude's creator with 10x leverage
Tokens
DeFi
NFTs
Network: Arc
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app.remix.live
Account 1
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Money
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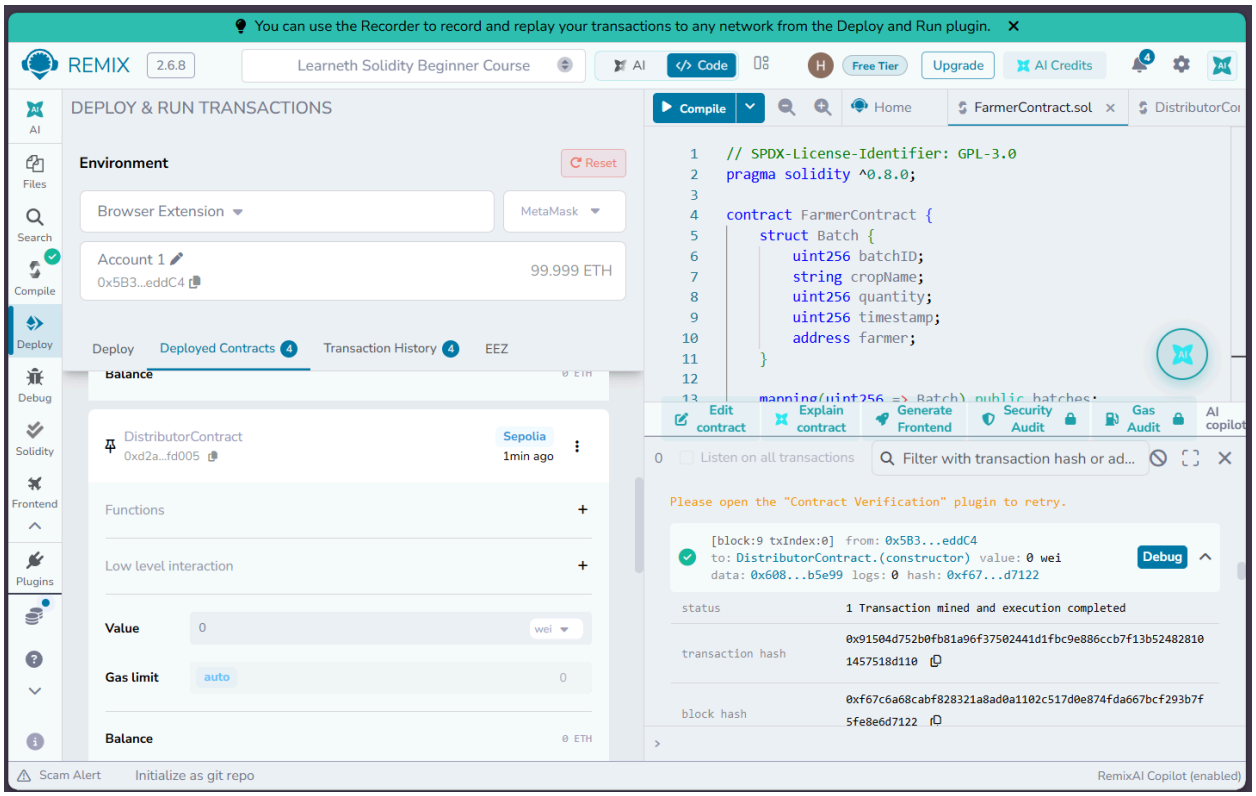

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AI
Code
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Upgrade
AI Credits
SOLIDITY COMPILER
COMPILER
0.8.18+commit.87f61d96
Include nightly builds
Auto compile
Hide warnings
Advanced Configurations
Compiler configuration
LANGUAGE
Solidity
EVM VERSION
paris
Optimization
200
Use configuration file
Load AI Skills
Compile CustomerContract....
Security Audit
Gas Audit
Compile
ibutorContract.sol
RetailerContract.sol
CustomerContract.sol
// SPDX-License-Identifier: GPL-3.0
pragma solidity ^0.8.0;
interface IRetailerContract {
function getRetailBatch(uint256 _batchID) - gas
external
view
returns (
uint256,
string memory,
uint256,
uint256,
address,
address,
address
);
}
contract CustomerContract {
struct PurchasedBatch {
uint256 batchID;
string cropName;
uint256 quantity;
Edit contract
Explain contract
Generate Frontend
Security Audit
Gas Audit
AI copilot
Listen on all transactions
Filter with transaction hash or address
Type the library name to see available commands.
Scam Alert
Initialize as git repo
RemixAI Copilot (enabled)

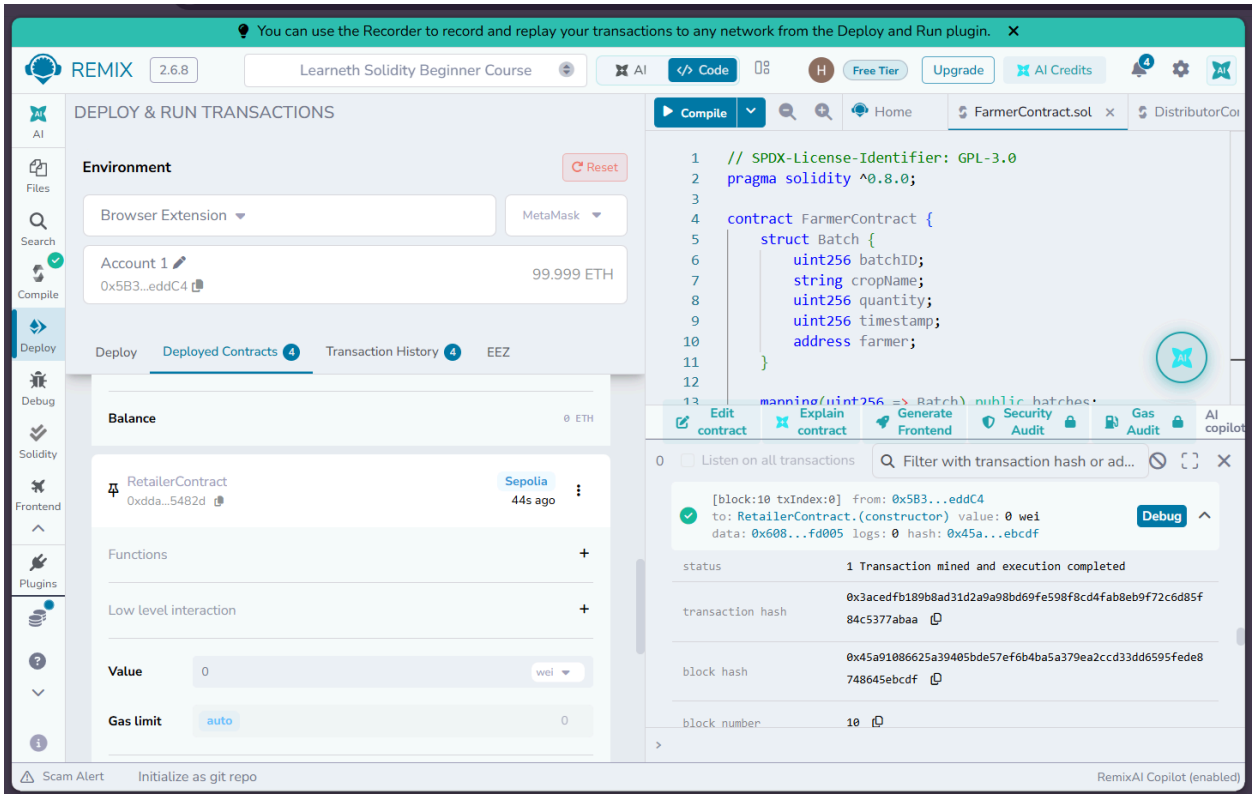

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Include nightly builds
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Use configuration file
Load AI Skills
Compile RetailerContract.sol
Security Audit
Gas Audit
Compile
ibutorContract.sol
RetailerContract.sol
CustomerContract.sol
// SPDX-License-Identifier: GPL-3.0
pragma solidity ^0.8.0;
interface IDistributorContract {
function getReceivedBatch(uint256 _batchID) - gas
external
view
returns (
uint256,
string memory,
uint256,
uint256,
address,
address
);
}
contract RetailerContract {
struct RetailBatch {
uint256 batchID;
string cropName;
uint256 quantity;
uint256 timestamp;
Edit contract
Explain contract
Generate Frontend
Security Audit
Gas Audit
AI copilot
Listen on all transactions
Filter with transaction hash or address
Type the library name to see available commands.
Scam Alert
Initialize as git repo
RemixAI Copilot (enabled)

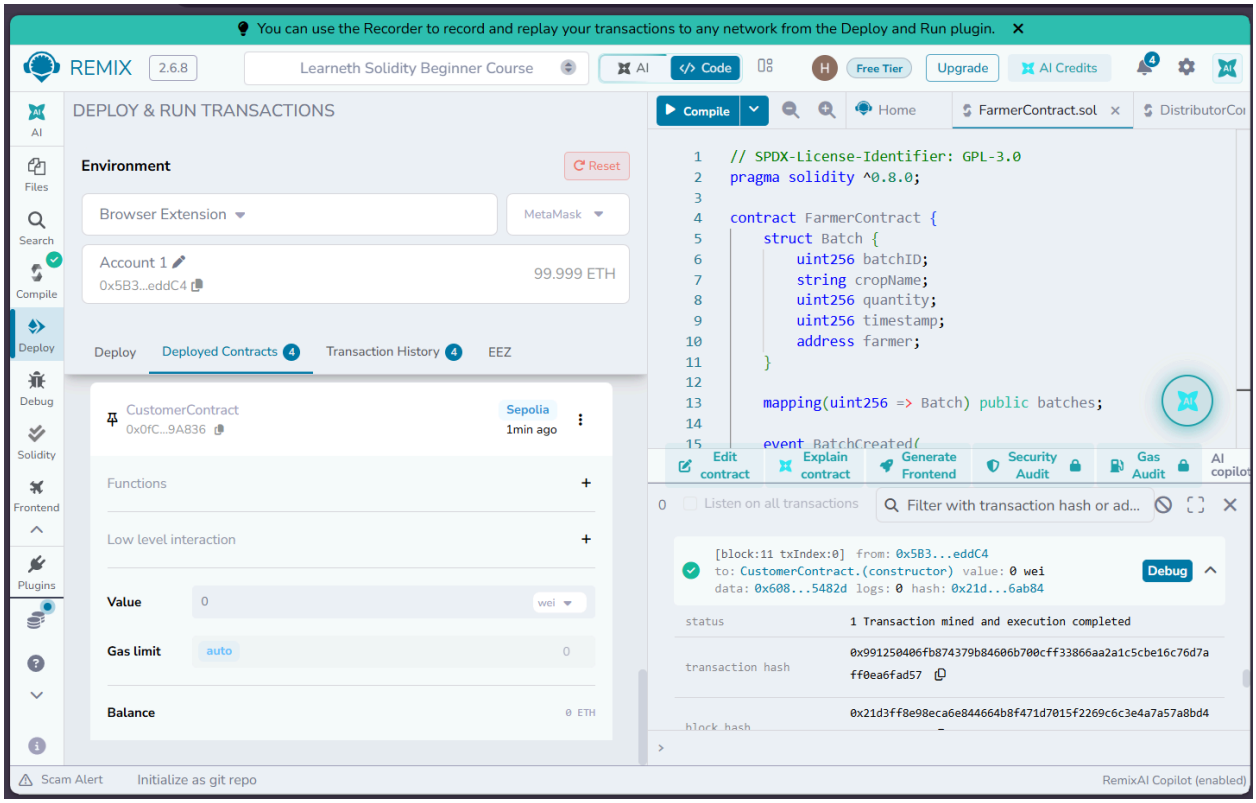

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Include nightly builds
Auto compile
Hide warnings
Advanced Configurations
Compiler configuration
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Optimization
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Use configuration file
Load AI Skills
Compile DistributorContract...
Security Audit
Gas Audit
Compile
DistributorContract.sol
RetailerContract.sol
CustomerContract.sol
// SPDX-License-Identifier: GPL-3.0
pragma solidity ^0.8.0;
interface IFarmerContract {
function getBatch(uint256 _batchID) - gas
external
view
returns (
uint256,
string memory,
uint256,
uint256,
address
);
}
contract DistributorContract {
struct ReceivedBatch {
uint256 batchID;
string cropName;
uint256 quantity;
uint256 timestamp;
address farmer;
Edit contract
Explain contract
Generate Frontend
Security Audit
Gas Audit
AI copilot
Listen on all transactions
Filter with transaction hash or address
Type the library name to see available commands.
Scam Alert
Initialize as git repo
RemixAI Copilot (enabled)

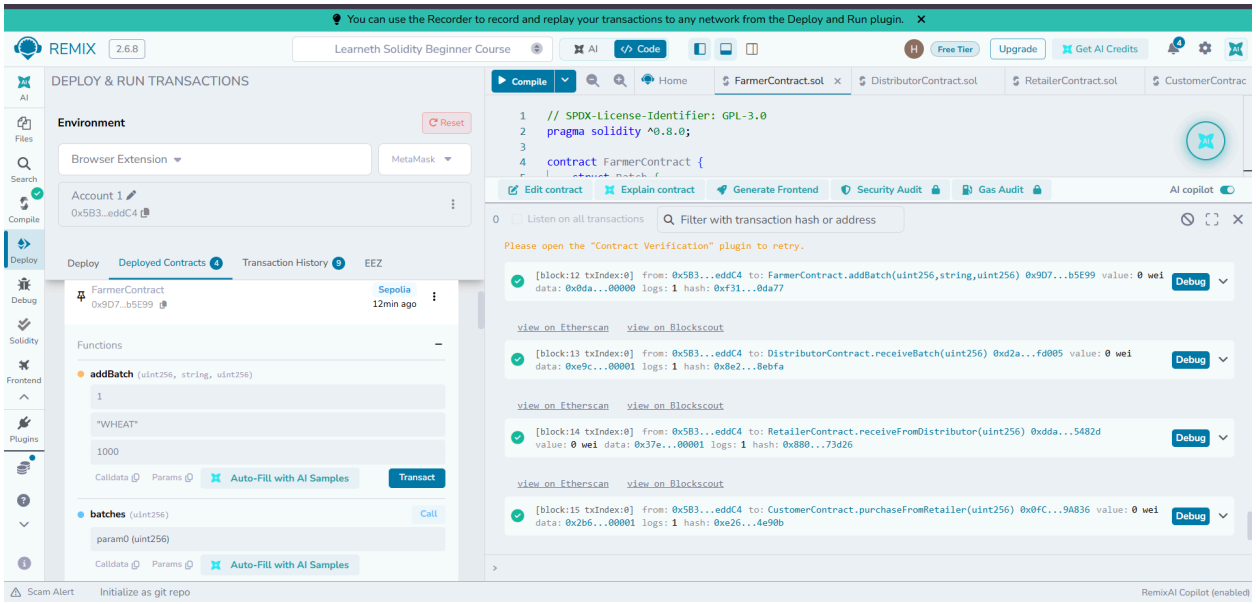

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Include nightly builds
Auto compile
Hide warnings
Advanced Configurations
Compiler configuration
LANGUAGE
Solidity
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Optimization
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Use configuration file
Load AI Skills
Ctrl+S to compile FarmerContract.sol
Compile FarmerContract.sol
Security Audit
Gas Audit
Compile
Home
FarmerContract.sol
DistributorContract.sol
RetailerContract.so
// SPDX-License-Identifier: GPL-3.0
pragma solidity ^0.8.0;
contract FarmerContract {
struct Batch {
uint256 batchID;
string cropName;
uint256 quantity;
uint256 timestamp;
address farmer;
}
mapping(uint256 => Batch) public batches;
event BatchCreated(
uint256 batchID,
string cropName,
uint256 quantity,
uint256 timestamp,
address farmer
);
function addBatch(infinite gas
Edit contract
Explain contract
Generate Frontend
Security Audit
Gas Audit
AI copilot
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Filter with transaction hash or address
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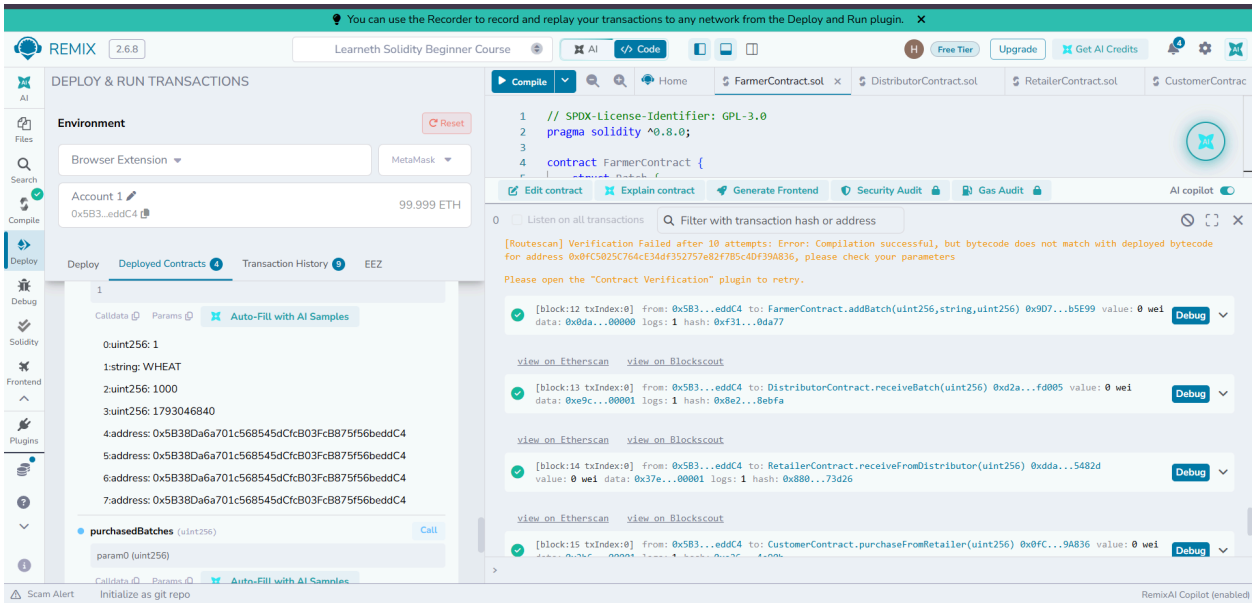

You can use the Recorder to record and replay your transactions to any network from the Deploy and Run plugin.
REMIX 2.6.8
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AI
Code
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AI Credits
DEPLOY & RUN TRANSACTIONS
Environment
Reset
Browser Extension
MetaMask
Account 1
0x5B3...eddC4
99.999 ETH
Deploy
Deployed Contracts 4
Transaction History 4
EEZ
FarmerContract
0x9D7...b5E99
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Functions
Low level interaction
Value
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wei
Gas limit
auto
0
Balance
0 ETH
Compile
Home
FarmerContract.sol
DistributorCor
// SPDX-License-Identifier: GPL-3.0
pragma solidity ^0.8.0;
contract FarmerContract {
struct Batch {
uint256 batchID;
string cropName;
uint256 quantity;
uint256 timestamp;
address farmer;
}
Edit contract
Explain contract
Generate Frontend
Security Audit
Gas Audit
AI copilot
Listen on all transactions
Filter with transaction hash or ad...
[block:8 txIndex:0] from: 0x5B3...eddC4
to: FarmerContract.(constructor) value: 0 wei
data: 0x608...20033 logs: 0 hash: 0x0c3...c9da4
Debug
status
1 Transaction mined and execution completed
transaction hash
0xf875bca0f25470059480945c20a1621beb963902b5a4db0392dcbc0da322ab08
block hash
0x0c3cebe21b4425240e98afc39509829d01f235799c02708622b053e6141c9da4
Scam Alert
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RemixAI Copilot (enabled)


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Environment
Reset
Browser Extension
MetaMask
Account 1
0x5B3...eddC4
99.999 ETH
Deploy
Deployed Contracts 4
Transaction History 4
EEZ
Balance
0 ETH
DistributorContract
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Gas limit
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0
Balance
0 ETH
Compile
Home
FarmerContract.sol
DistributorCor
// SPDX-License-Identifier: GPL-3.0
pragma solidity ^0.8.0;
contract FarmerContract {
struct Batch {
uint256 batchID;
string cropName;
uint256 quantity;
uint256 timestamp;
address farmer;
}
Edit contract
Explain contract
Generate Frontend
Security Audit
Gas Audit
AI copilot
Listen on all transactions
Filter with transaction hash or ad...
Please open the "Contract Verification" plugin to retry.
[block:9 txIndex:0] from: 0x5B3...eddC4
to: DistributorContract.(constructor) value: 0 wei
data: 0x608...b5e99 logs: 0 hash: 0xf67...d7122
Debug
status
1 Transaction mined and execution completed
transaction hash
0x91504d752b0fb81a96f37502441d1fbc9e886ccb7f13b52482810457518d110
block hash
0xf67c6a68cabf828321a8ad0a1102c517d0e874fda667bcf293b7f5fe8e6d7122
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RemixAI Copilot (enabled)


You can use the Recorder to record and replay your transactions to any network from the Deploy and Run plugin.
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AI
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DEPLOY & RUN TRANSACTIONS
Environment
Reset
Browser Extension
MetaMask
Account 1
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Deployed Contracts 4
Transaction History 4
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RetailerContract
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44s ago
Functions
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Value
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wei
Gas limit
auto
0
Compile
Home
FarmerContract.sol
DistributorCo
// SPDX-License-Identifier: GPL-3.0
pragma solidity ^0.8.0;
contract FarmerContract {
struct Batch {
uint256 batchID;
string cropName;
uint256 quantity;
uint256 timestamp;
address farmer;
}
mapping(uint256 => Batch) public batches;
Edit contract
Explain contract
Generate Frontend
Security Audit
Gas Audit
AI copilot
Listen on all transactions
Filter with transaction hash or ad...
[block:10 txIndex:0] from: 0x5B3...eddC4
to: RetailerContract.(constructor) value: 0 wei
data: 0x608...fd005 logs: 0 hash: 0x45a...ebcdf
Debug
status
1 Transaction mined and execution completed
transaction hash
0x3acedfb189b8ad31d2a9a98bd69fe598f8cd4fab8eb9f72c6d85f84c5377abaa
block hash
0x45a91086625a39405bde57ef6b4ba5a379ea2ccd33dd6595fede8748645ebcdf
block number
10
Scam Alert
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RemixAI Copilot (enabled)


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DEPLOY & RUN TRANSACTIONS
Environment
Reset
Browser Extension
MetaMask
Account 1
0x5B3...eddC4
99.999 ETH
Deploy
Deployed Contracts 4
Transaction History 4
EEZ
Balance
0 ETH
CustomerContract
0x0fC...9A836
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1min ago
Functions
Low level interaction
Value
0
wei
Gas limit
auto
0
Compile
Home
FarmerContract.sol
DistributorCo
// SPDX-License-Identifier: GPL-3.0
pragma solidity ^0.8.0;
contract FarmerContract {
struct Batch {
uint256 batchID;
string cropName;
uint256 quantity;
uint256 timestamp;
address farmer;
}
mapping(uint256 => Batch) public batches;
event BatchCreated(
Edit contract
Explain contract
Generate Frontend
Security Audit
Gas Audit
AI copilot
Listen on all transactions
Filter with transaction hash or ad...
[block:11 txIndex:0] from: 0x5B3...eddC4
to: CustomerContract.(constructor) value: 0 wei
data: 0x608...5482d logs: 0 hash: 0x21d...6ab84
Debug
status
1 Transaction mined and execution completed
transaction hash
0x991250406fb874379b84606b700cff33866aa2a1c5cbe16c76d7aff0ea6fad57
0x21d3ff8e98eca6e844664b8f471d7015f2269c6c3e4a7a57a8bd4
Scam Alert
Initialize as git repo
RemixAI Copilot (enabled)





Conclusion :

Thus, the experiment provides practical understanding of Solidity program structure, basic constructs, smart contract compilation, deployment, transactions, and interaction with contracts. The Solidity Basics course and its assignments help demonstrate these concepts through hands-on smart contract development.